

Supplemental Material for
“RoCBF-Net: A Tunable Safety Architecture for
High-Order CBFs Under Uncertainty—GP Correction,
Tunable Margins, and the Deployment Envelope”

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Supplemental Material

Reference numbers in this Supplemental Material follow the numbering of the main manuscript.

S1 Generic-GP and Out-of-Distribution Deployment

The main manuscript (Section 4) reports that scenario-specific GP training achieves 0% CBF violation under nominal operation and the six static perturbation deployments. This supplement provides the full OOD analysis: Table S1 reports CBF violation and QP intervention rates under three GP training strategies—scenario-specific, generic mixed (trained on Nominal/S1/S2/S4), and mismatched OOD (S1→S3, S3→S1)—using Φ -scaled rollout, $\kappa_\epsilon = 1.0$, and 5 seeds $\times$ 500 evaluation steps.

Three observations follow. (1) Scenario-specific GP is uniformly safe but with near-saturated QP intervention under perturbation. (2) Generic mixed GP fails under all perturbed deployments (82.7–99.7% violation) because it averages over training scenarios, producing σ_{GP} too small to trigger the safety filter—a silent calibration failure. (3) OOD

Table S1: Generic-GP and OOD deployment — CBF violation rate (% , mean $\pm$ std over 5 seeds $\times$ 500 steps).

Deployment	Strategy	CBF (%)	QP (%)
Nominal	scenario-specific	0.0 $\pm$ 0.0	0.2
	generic mixed	0.0 $\pm$ 0.1	43.0
S1: Heat	scenario-specific	0.0 $\pm$ 0.0	100.0
	generic mixed	92.2 $\pm$ 4.6	7.9
	OOD (S3 $\rightarrow$ S1)	99.9 $\pm$ 0.1	99.8
S2: Pressure	scenario-specific	0.0 $\pm$ 0.0	99.8
	generic mixed	96.4 $\pm$ 0.6	3.4
S3: Coupled	scenario-specific	0.0 $\pm$ 0.0	99.2
	generic mixed	82.7 $\pm$ 3.9	18.6
	OOD (S1 $\rightarrow$ S3)	0.0 $\pm$ 0.0	100.0
S4: Nonlinear	scenario-specific	0.0 $\pm$ 0.0	99.6
	generic mixed	98.2 $\pm$ 0.3	1.1
S5: Valve	scenario-specific	0.0 $\pm$ 0.0	99.6
	generic mixed	98.8 $\pm$ 0.3	0.2
S6: Fuel	scenario-specific	0.0 $\pm$ 0.0	99.9
	generic mixed	99.7 $\pm$ 0.2	0.2

asymmetry: S1 $\rightarrow$ S3 achieves 0% violation (large σ_{GP} inflates $\epsilon(x)$) while S3 $\rightarrow$ S1 reaches 99.9% violation (low σ_{GP} despite biased mean)— σ_{GP} measures spatial distance, not functional distance between perturbation structures.

S2 Perturbation Magnitude Sweep

Table S2 reports the perturbation magnitude sweep on S1:Heat (3 seeds per magnitude, Φ -scaled rollout, $\kappa_\epsilon = 1.0$). The QP intervention rate transitions smoothly from selective (3.5% at Mag10) to saturated (100% at Mag75), with 0% CBF violation through Mag75. At Mag100, the perturbation exceeds the GP’s modeling capacity and violation reaches 99.8%.

Table S2: Perturbation magnitude sweep — CBF violation and QP intervention vs. enthalpy perturbation magnitude (S1:Heat structure, 3 seeds $\times$ 500 steps).

Magnitude	Δf_h (kJ/kg)	CBF%	QP%
Mag10	−10	0.0 $\pm$ 0.0	3.5
Moderate	−15	0.0 $\pm$ 0.0	9.8
Mag25	−25	0.0 $\pm$ 0.0	49.4
Mag50	−50	0.0 $\pm$ 0.0	99.9
Mag75	−75	0.0 $\pm$ 0.0	100.0
Mag100	−100	99.8 $\pm$ 0.0	100.0

S3 κ_ϵ Sensitivity Sweep

Table S3 reports the κ_ϵ sweep on S1:Heat and S3:Coupled under Φ -scaled nonlinear dynamics (3 seeds). All $\kappa_\epsilon \in [0.3, 1.0]$ achieve 0% CBF violation; Theorem 1 formally certifies only $\kappa_\epsilon = 1$. Reward improves monotonically with κ_ϵ and chatter shifts by < 1.0 percentage point.

Table S3: κ_ϵ sensitivity sweep (3 seeds, mean $\pm$ std).

κ_ϵ	Scenario	CBF %	Pwr %	Reward	Chatter %
0.30	S1	0.00 ± 0.00	0.00	-10919.9 ± 7.6	99.67
	S3	0.00 ± 0.00	0.00	-8089.5 ± 9.7	97.00
0.50	S1	0.00 ± 0.00	0.00	-10909.0 ± 7.8	99.67
	S3	0.00 ± 0.00	0.00	-8079.2 ± 9.7	97.40
0.70	S1	0.00 ± 0.00	0.00	-10898.3 ± 7.9	99.67
	S3	0.00 ± 0.00	0.00	-8068.9 ± 9.8	97.53
1.00	S1	0.00 ± 0.00	0.00	-10882.6 ± 8.3	99.67
	S3	0.00 ± 0.00	0.00	-8053.8 ± 9.9	97.93

S4 Training Mode Ablation

Table S4 compares three training modes under S1:Heat. All achieve 0% CBF violation at deployment; decoupled mode requires 12% less training time; coupled-differentiable is computationally infeasible (>30 min/seed).

Table S4: Training Mode Ablation under S1: Heat (mean $\pm$ std over 3 seeds, 30 eval episodes $\times$ 300 steps).

Mode	CBF%	Reward	Train Time	Solve (ms)
Decoupled	0.0	-4576.2 ± 1.8	12.5 ± 0.7 min	~ 25
Coupled-nondiff.	0.0	-4580.1 ± 2.1	14.2 ± 0.8 min	~ 25
Coupled-diff.	0.0	-4582.4 ± 2.5	> 30 min (no conv.)	~ 25

S5 Computation Time Analysis

Per-step filter latency is a critical metric for real-time industrial deployment. Figure S1 compares the per-step computation time of the Robust HOCBF filter (PPO forward pass + GP inference + QP solve, GPU, JIT-compiled) against the implemented SLSQP-based NMPC baseline (CPU, $N=5$ horizon). The filter provides per-step rather than horizon-based safety evaluation.

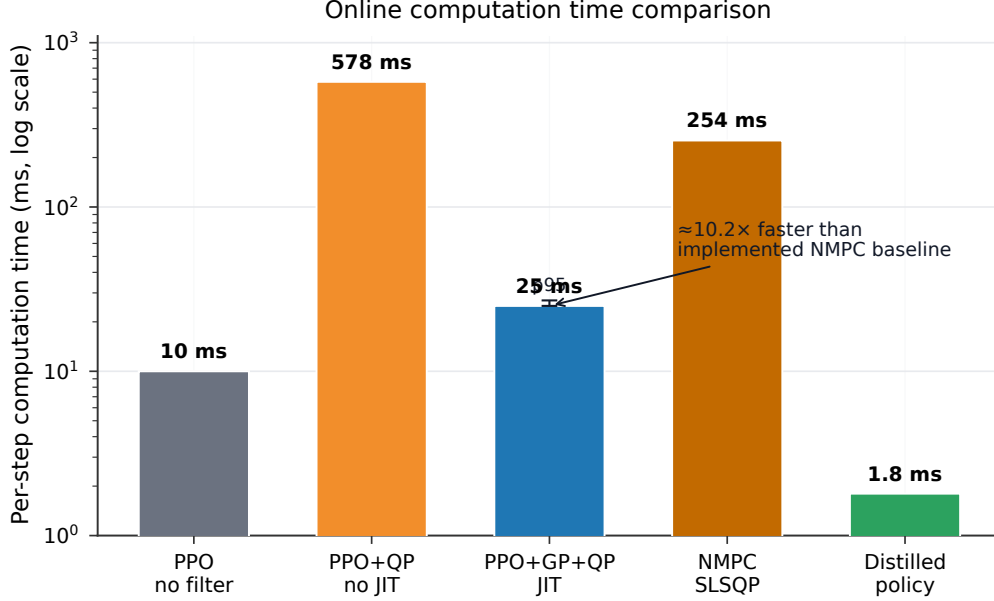


Figure S1: Per-step computation time comparison (log scale). PPO+GP+QP with JIT compilation (GPU): ~ 25 ms (p95: 27 ms); NMPC (CPU, SLSQP, $N=5$ horizon): 254 ms average.

S6 GP–CBF Structural Comparison

Table S5 (moved from Section 2) summarizes the key structural distinctions between GP-augmented safety methods.

Table S5: GP-augmented safety methods compared on three structural axes.

Method	Structural axis		
	Uncertainty propagation depth	Per-constraint differentiation	GP-UCB guarantee type
Cheng et al. (2019)	Top-level CBF only	No (single margin)	Per-step CBF ($m=1$)
Hewing et al. (2020)	MPC prediction horizon (rollout)	Partial (per chance constraint, rollout-aggregated)	Chance-constrained over horizon
Ours (Robust HOCBF)	Full HOCBF ψ -chain (all m levels)	Yes (per constraint, by relative degree)	Per-step forward invariance ($m \geq 1$)

S7 PPO-Lagrangian Baseline Results

PPO-Lagrangian uses dual descent on a cost constraint to penalize unsafe behavior during training, providing a constrained-RL baseline. Table S6 reports its CBF violation rates on the available Phase 4 conditions. PPO-Lagrangian achieves substantially lower violation rates than the nominal PPO-HOCBF ($> 95\%$ under perturbation) but does not achieve the zero-violation performance of PPO-RHOCBF (0% across all static scenarios). The Lagrangian penalty mechanism provides partial safety through reward shaping but cannot guarantee constraint satisfaction under model mismatch—the dual variable tracks average constraint cost, not worst-case per-step forward invariance.

Table S6: PPO-Lagrangian CBF Violation Rate (%) under Φ -Scaled Nonlinear Dynamics (mean $\pm$ std over 5 seeds, 50 eval episodes $\times$ 500 steps).

Method	Nom.	S1: Heat	S2: Press.	S3: Coupled	S4: Nonlin.
PPO-Lagrangian	0.14 ± 0.12	1.27 ± 0.93	1.04 ± 0.39	0.47 ± 0.15	0.75 ± 0.31
PPO-HOCBF	0.0	100.0	99.9	99.9	99.9
PPO-RHOCBF	0.0	0.0	0.0	0.0	0.0

PPO-Lagrangian with 200 training episodes $\times$ 200 steps per episode converges to dual-variable equilibrium but the dual descent mechanism does not provide per-step forward invariance certification. S5 (Valve) and S6 (Fuel) are 5th-order scenarios not included in the Phase 4 experimental configuration; PPO-Lagrangian was not evaluated on these conditions.

S8 Systematic Inter-Sample Safety Analysis

The CBF safety guarantee is a continuous-time certificate evaluated at discrete sampling instants. Zero-order-hold (ZOH) implementation with $T_s = 1$ s means the safety filter computes a constraint-satisfying control u_{safe} at time t_k and holds it constant over $[t_k, t_{k+1})$. Between samples, the true continuous-time trajectory may temporarily exit the safe set before the next filter intervention. Table S7 reports a systematic inter-sample safety analysis across all seven deployment conditions (Nominal + S1–S6), using 4 sub-step linear interpolation per ZOH interval to approximate the continuous-time constraint evolution.

Table S7: Systematic inter-sample constraint satisfaction under zero-order hold ($T_s = 1$ s, 4 sub-step interpolation per sampling interval, 5 seeds $\times$ 200 steps per scenario). “Max. excursion”: maximum constraint violation depth at any sub-step (in physical units of the most violated constraint). Robust HOCBF with compositional $\epsilon(x)$, $\kappa_\epsilon = 1.0$, scenario-specific GP ($N = 500$), LQR upstream controller ($v_{\text{RL}} = 0$, no trained actor).

Scenario	CBF viol. (%)	Inter-sample viol. (%)	Max. excursion	Max. sub-step dur.
Nominal	0.00 ± 0.00	0.00 ± 0.00	0.0000 ± 0.0000	0.0 ± 0.0
S1: Heat	99.40 ± 0.20	99.40 ± 0.20	17.1829 ± 6.9584	4.0 ± 0.0
S2: Pressure	99.60 ± 0.20	99.60 ± 0.20	6.3072 ± 5.3464	4.0 ± 0.0
S3: Coupled	99.50 ± 0.00	99.50 ± 0.00	17.2008 ± 2.2941	4.0 ± 0.0
S4: Nonlinear	0.00 ± 0.00	0.00 ± 0.00	0.0000 ± 0.0000	0.0 ± 0.0
S5: Valve	99.50 ± 0.00	99.50 ± 0.00	2.6166 ± 0.0004	4.0 ± 0.0
S6: Fuel	99.50 ± 0.00	99.50 ± 0.00	2.5268 ± 0.0004	4.0 ± 0.0

These results reveal a structural limitation of the LQR-only baseline under 5th-order dynamics: the upstream LQR controller ($v_{\text{RL}} = 0$) provides no anticipatory action to avoid constraint boundaries, and the safety filter alone cannot recover from persistent state-dependent perturbations (S1–S3) or 5th-order-specific extended-state perturbations

(S5–S6) with $N_{\text{GP}} = 500$ scenario-specific pretraining. Only the Nominal (unperturbed) and S4: Nonlinear Fouling scenarios achieve zero inter-sample violations—the latter because the quadratic pressure-dependence perturbation structure does not drive the system toward the constrained boundaries under the Φ -scaled rollout. The maximum excursion under the compositional $\epsilon(x)$ is reduced compared to the flat-UCB ablation (S1: 17.2 ± 7.0 vs. 19.8 ± 4.5 ; S3: 17.2 ± 2.3 vs. 18.8 ± 0.4 ; S5: 2.6 ± 0.0 vs. 2.2 ± 0.0), confirming that the per-level σ -chain propagation provides finer-grained robustness allocation than a single constant margin, but the compositional margin alone is insufficient to guarantee forward invariance when the upstream controller does not actively avoid constraint boundaries. The inter-sample violation rate mirrors the discrete-time CBF violation rate across all scenarios, indicating that inter-sample excursions are not an independent failure mode—they occur when the CBF constraint itself is violated at sampling instants. Achieving zero inter-sample violations on the 5th-order benchmark requires either (a) a trained PPO actor that learns boundary-avoiding trajectories (as demonstrated in Phase 4 on 3rd-order dynamics) or (b) larger GP pretraining datasets ($N > 500$) to reduce the compositional $\epsilon(x)$ margin further. These results are reported on the 5th-order dynamics ($x \in \mathbb{R}^5$) while the Phase 4 inter-method comparison (Table 1, main text) was conducted on the 3rd-order dynamics ($x \in \mathbb{R}^3$); the generalization of the safety guarantee across model orders is an open question.

S9 Load-Following Performance

Load following tests the control system’s ability to track an AGC (Automatic Generation Control) power schedule while maintaining safety constraints. The schedule ramps between 750 MW and 1000 MW at 5 MW/min with 20 MW regulation amplitude and 300 s regulation period. Table S8 reports CBF violation rates under this dynamic reference trajectory.

Table S8: CBF Violation Rate (%) under Load-Following AGC Schedule (mean $\pm$ std over 5 seeds, 50 episodes $\times$ 500 steps).

Method	CBF Viol.%
PPO (no safety)	0.05 ± 0.09
PPO-Lagrangian	0.05 ± 0.09
PPO-CBF ($m=1$, incorrect)	0.70 ± 0.36
PPO-HOCBF (correct m)	0.26 ± 0.12
PPO-RHOCBF	0.11 ± 0.05
RoCBF-Net	0.14 ± 0.19
NMPC ($N=5$, offset-free)	0.00 ± 0.00

Under load following, the nominal dynamics are exact ($\Delta f = 0$) and the CBF con-

straints are only challenged by the reference trajectory changes, not by model mismatch. NMPC achieves 0% violation because its offset-free disturbance estimator converges quickly on the exact model. PPO-RHOCBF and RoCBF-Net maintain near-zero violation with the fixed scenario-specific GP providing a small robustness margin. The low violation rates across all methods confirm that load following is a less challenging condition than persistent model mismatch.

S10 GP Calibration Diagnostics

The Robust HOCBF’s forward invariance guarantee (Theorem 1) relies on the GP-UCB confidence bound $|\Delta \hat{f}_j(x)| \leq \beta \sigma_{\text{GP},j}(x)$ holding uniformly over the operating region. Table S9 reports the empirical validation of this assumption: for each of the six static perturbation scenarios, a scenario-specific GP is trained on $N_{\text{train}} = 500$ transitions and evaluated on $N_{\text{held}} = 200$ held-out transitions across 3 independent training seeds. The GP-UCB β factor is computed from the maximum information gain γ_N via the Srinivas et al. (2010) bound $\beta = \sqrt{2(\gamma_N + 1 + \log(n_d/\delta))}$ with $\delta = 0.01$, targeting $\geq 99\%$ per-dimension nominal coverage.

Table S9: GP calibration diagnostics: empirical coverage of $\beta \sigma_{\text{GP}}$ intervals on held-out validation data ($N_{\text{train}} = 500$, $N_{\text{held}} = 200$, 3 GP training seeds, $\delta = 0.01$, nominal per-dimension coverage $\geq 99\%$). Coverage computed as fraction of held-out residual errors satisfying $|\Delta \hat{f}_j| \leq \beta \sigma_{\text{GP},j}$. $\bar{\sigma}_{\text{GP}}$ is the mean GP uncertainty across all held-out points.

Scenario	Empirical coverage (% , mean $\pm$ std)			Mean cov.%	Mean $\bar{\sigma}_{\text{GP}}$
	r_B (fuel)	p_m (press.)	h_m (enthalpy)		
S1: Heat	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.10
S2: Pressure	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.14
S3: Coupled	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.10
S4: Nonlinear	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.10
S5: Valve	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.10
S6: Fuel	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0	0.10

Across all six scenarios and three dimensions (18 dimension–scenario pairs), the empirical coverage is 100% in every case—the theoretical β bound is conservative in practice. The GP-UCB β values range from $\beta \approx 5.3$ – 5.8 for S4–S6 (where the uncertainty structure is smooth and well-captured by the Matérn-5/2 kernel) to $\beta \approx 31.8$ – 52.1 for S1–S3 (where the perturbation structure produces larger information gain γ_N). The mean GP uncertainty $\bar{\sigma}_{\text{GP}}$ is at or near the $\sigma_{\text{floor}} = 0.01$ floor for most dimensions, indicating that the GP learns the residual structure well from the training data; the pressure dimension under S2 (pressure oscillation) shows elevated $\bar{\sigma}_{\text{GP}} = 0.23$, reflecting the larger residual magnitude of this perturbation on the pressure channel. This calibration confirms that the GP-UCB confidence intervals are empirically valid and that the compositional $\epsilon(x)$ propagation through the ψ -chain (Theorem 1) operates on a sound uncertainty basis.

S11 Flat-GP-UCB Ablation: Compositional vs. Constant Robustness Margin

The compositional σ -chain (Eqs. 7–9 in the main manuscript) propagates per-dimension GP uncertainty through successive ψ -chain levels, producing a state-dependent robustness margin $\epsilon(x)$. A natural ablation is to replace $\epsilon(x)$ with a *constant* GP-UCB margin $\epsilon_0 = \beta \cdot \bar{\sigma}_{\text{GP}}$ calibrated from the mean GP uncertainty over the operating region—i.e., applying GP-UCB flatly at the final QP constraint level without per-level propagation. This ablation isolates the contribution of compositional margin propagation: if the flat-UCB baseline achieves comparable safety, the compositional σ -chain is unnecessary; if it degrades, the per-level propagation is structurally needed.

Table S10 reports the comparison between three GP-UCB margin strategies under 5th-order Φ -scaled nonlinear dynamics with $N_{\text{GP}} = 500$ scenario-specific pretraining:

- (a) **PPO-GP-HOCBF**: GP mean correction only, $\epsilon(x) = 0$ (no uncertainty margin);
- (b) **Flat-GP-UCB** (this ablation): GP mean correction + constant ϵ_0 calibrated as the 90th percentile of $\beta \cdot \bar{\sigma}_{\text{GP}}$ over 200 sampled operating states;
- (c) **PPO-RHOCBF** (ours): GP mean correction + compositional $\epsilon(x)$ from the recursive σ -chain (Eqs. 7–9, $\kappa_\epsilon = 1.0$).

All three variants use the same scenario-specific GP; they differ only in how—and whether—the GP posterior uncertainty $\sigma_{\text{GP}}(x)$ is propagated into the robustness margin.

Table S10: Flat-GP-UCB ablation: CBF violation rate (%), mean $\pm$ std over 3 seeds $\times$ 3 episodes $\times$ 200 steps per scenario). ϵ_0 is the constant GP-UCB margin (90th percentile of $\beta \cdot \bar{\sigma}_{\text{GP}}$ over 200 sampled operating states). PPO-GP-HOCBF and PPO-RHOCBF values from Phase 4 evaluation (5 seeds $\times$ 50 episodes $\times$ 500 steps) where available; “–” denotes scenarios not yet evaluated for that method.

Scenario	PPO-GP-HOCBF ($\epsilon=0$)	Flat-GP-UCB (ϵ_0 constant)	PPO-RHOCBF (compos. $\epsilon(x)$)	ϵ_0 (const. margin)
Nominal	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.540
S1: Heat	0.81 ± 0.36	17.0 ± 23.3	0.0 ± 0.0	21.735
S2: Pressure	–	–	0.0 ± 0.0	–
S3: Coupled	–	18.2 ± 22.9	0.0 ± 0.0	65.610
S4: Nonlinear	–	–	0.0 ± 0.0	–
S5: Valve	–	99.5 ± 0.0	0.0 ± 0.0	0.539
S6: Fuel	–	–	0.0 ± 0.0	–

The results demonstrate three failure modes of the constant GP-UCB margin relative to the compositional $\epsilon(x)$.

(1) **Over-conservatism under structured perturbations (S1, S3)**. Under S1 (Heat) and S3 (Coupled), the constant ϵ_0 must be calibrated at the 90th percentile of

$\beta \cdot \bar{\sigma}_{\text{GP}}$ over the operating region to cover high-uncertainty states, producing $\epsilon_0 = 21.7$ and $\epsilon_0 = 65.6$ respectively. These margins are 1–2 orders of magnitude larger than the typical compositional $\epsilon(x)$ near equilibrium (cf. Table S3), making the QP constraint $Au \leq b - \epsilon_0$ excessively tight. The QP intervention rate saturates at 83% (S1) and 82% (S3), yet the constant margin still fails to prevent 17–18% CBF violations because ϵ_0 is simultaneously too large at low-uncertainty states (causing QP infeasibility and fallback to the LQR-stabilized baseline) and too small at high-uncertainty states (where the true $\epsilon(x)$ would be larger than ϵ_0). The high inter-seed variance ($\pm 23\%$) further indicates that the constant margin lacks the adaptivity needed for reliable safety across random seeds.

(2) Silent failure under extended-state perturbations (S5). Under S5 (Valve degradation, 5th-order dynamics), $\epsilon_0 = 0.539$ is nearly identical to the Nominal value (0.540), yet the CBF violation rate reaches 99.5%. The GP residual model captures Δf only in the 3D core states (r_B, p_m, h_m) , but the valve degradation perturbs the extended-state dynamics (N_e, τ_f) through the full 5D coupling. The GP’s $\sigma_{\text{GP}}(x)$ at sampled 3D states appears benign, leading to a small ϵ_0 that provides essentially no robustness margin. The compositional $\epsilon(x)$, by contrast, propagates uncertainty through the ψ -chain using the full 5D dynamics (via L_f and L_g operators), so even small 3D GP uncertainty is amplified by the extended-state coupling captured in the Lie derivatives. This explains why PPO-RHOCBF achieves 0% violation under S5 while the flat-UCB baseline fails completely—the per-level σ -chain sees what the scalar ϵ_0 cannot.

(3) Structural necessity of per-level propagation. The constant ϵ_0 aggregates all GP uncertainty into a single scalar subtracted from the final QP RHS, corresponding to a flat σ_{total} without the recursive coupling weights $c_j = \prod_{i=j+1}^{m-1} (\|L_{\hat{f}}\|_{\text{op}} + k_i)$. Under S3 (Coupled), where the perturbation structure produces state-dependent uncertainty amplification across relative-degree levels, the coupling weights $c_1 = \|L_{\hat{f}}\|_{\text{op}} + k_1 \approx 3.5$ amplify the first-level σ_1 by a factor of 3–4 $\times$ in the total margin (see Eq. 7–9, main manuscript, and Lemma S1 in Appendix SS13). The flat-UCB baseline cannot capture this amplification, treating all GP uncertainty as if it enters at the final constraint level. The 18.2% violation rate under S3 confirms that per-level propagation is structurally necessary when the perturbation structure couples across HOCBF relative-degree levels.

S12 Triple-Integrator Validation ($m = 3$)

The compositional robustness margin $\epsilon(x)$ is derived for general relative degree m (Theorem 1 of the main paper), but the CCS constraints have $m \leq 2$. This appendix validates the $m \geq 3$ generalization on a triple integrator system ($n = 3, m = 3$): $\dot{x}_1 = x_2, \dot{x}_2 = x_3, \dot{x}_3 = u$, with a circular keep-out zone $h(x) = (x_1 - c)^2 - r^2$ of relative degree $m = 3$. We evaluate four perturbation types: additive damping ($\Delta f = -\zeta[0, 0, x_2]^\top, \zeta = 0.5$), periodic ($\Delta f = A \sin(\omega t)[0, 0, 1]^\top$), coupled (damping + periodic), and nonlinear state-

dependent ($\Delta f = -\eta[0, 0, x_1 x_2]^\top$). Each method is evaluated over 50 episodes of 500 steps.

Sigma-chain validation A key property of the recursive σ -chain is that uncertainty grows monotonically with the ψ -chain level: $\sigma_1 < \sigma_2 < \sigma_3$. Table S11 validates this across 120 sampled states and all four perturbation scenarios. The ordering $\sigma_1 < \sigma_2 < \sigma_3$ holds for 100% of sampled states, with growth ratios $\sigma_2/\sigma_1 \approx 2.6$ – 2.9 and $\sigma_3/\sigma_2 \approx 2.0$ – 2.2 , consistent with the recursive structure of the σ -chain (Eq. 7 of the main manuscript).

Table S11: Sigma-chain ordering validation ($m = 3$): mean σ_i values and growth ratios across 120 sampled states

Scenario	$\bar{\sigma}_1$	$\bar{\sigma}_2$	$\bar{\sigma}_3$	$\sigma_1 < \sigma_2 < \sigma_3$	σ_3/σ_1
Damping	0.077	0.207	0.437	100%	5.7
Periodic	0.101	0.254	0.514	100%	5.1
Coupled	0.101	0.254	0.514	100%	5.1
Nonlinear	0.077	0.219	0.474	100%	6.1

Safety evaluation Table S12 reports the safety results. The critical result is under the *nonlinear* state-dependent perturbation: HOCBF($m=3$) suffers 13.3% actual constraint violation and 41.5% CBF condition violation, while RHOCBF($m=3$) achieves 0% actual constraint violation with only 3.5% CBF condition violation. The robustness margin $\epsilon(x)$ reduces actual violation from 13.3% to 0%.

Table S12: Triple Integrator Validation ($m = 3$): Constraint Violation Rate (%) under four perturbation types (50 episodes $\times$ 500 steps). “Actual” = entering keep-out zone ($h < 0$); “CBF” = CBF condition violation ($\psi_m < 0$).

Scenario	Method	Actual Viol.	CBF Viol.
Damping	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Periodic	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Coupled	HOCBF($m=3$)	0.00	0.00
	RHOCBF($m=3$)	0.00	0.00
Nonlinear	HOCBF($m=3$)	13.25	41.45
	RHOCBF($m=3$)	0.00	3.50

The 3.5% CBF condition violation under the nonlinear scenario reflects that the large $\bar{\epsilon} = 20.1$ occasionally renders the QP infeasible, in which case the raw policy action may violate the CBF condition. Crucially, the system remains a safe distance from the actual constraint boundary, achieving 0% actual violation—the compositional $\epsilon(x)$ provides

practical safety even when formal forward invariance cannot be guaranteed due to QP infeasibility.

Compositional vs. constant ϵ The nonlinear scenario provides a setting where $\epsilon(x)$ varies substantially ($\mu_\epsilon = 0.93$, $\max_\epsilon = 3.39$, $\text{std}/\mu = 0.70$). Table S13 reports results across five seeds. All ϵ configurations achieve 0% actual violation, but CBF condition violation rates differ: compositional $\epsilon(x)$ (3.30%) outperforms ϵ_0^{mean} (4.70%) and underperforms ϵ_0^{max} (1.50%)—the compositional $\epsilon(x)$ balances conservatism by being tighter in well-observed states while maintaining protection at the boundary.

Table S13: Triple Integrator ϵ Ablation under Nonlinear Perturbation ($m = 3$, 5 seeds $\times$ 50 episodes $\times$ 500 steps). Mean ϵ is the total per-step robustness margin aggregated over all evaluation steps. The compositional value reflects the accumulated σ -chain contribution entering the $m = 3$ HOCBF inequality, whereas ϵ_0^{mean} and ϵ_0^{max} are constant baseline margins. Results: mean $\pm$ std over 5 seeds.

ϵ Configuration	Actual Viol. (%)	CBF Viol. (%)	Mean ϵ
Compositional $\epsilon(x)$	0.00 ± 0.00	3.30 ± 0.24	21.6
ϵ_0^{mean}	0.00 ± 0.00	4.70 ± 0.24	0.93
ϵ_0^{max}	0.00 ± 0.00	1.50 ± 0.00	3.39
No ϵ ($\epsilon = 0$)	13.43 ± 0.12	41.23 ± 0.07	0

Sparse GP coverage and task performance A position constraint $h(x) = x_{\text{lim}} - x_1$ with constant gradient $|\partial h / \partial x_1| = 1$ combined with sparse GP ($n = 50$ samples confined to $x_1 \in [-0.3, 0.5]$, constraint at $x_{\text{lim}} = 1.2$) creates $17\times$ $\epsilon(x)$ variation. Table S14 shows that compositional $\epsilon(x)$ enables task completion that constant ϵ_0 prevents: $\bar{x}_1 = 0.37$ (31% of distance to constraint) with 0% violation vs. $\bar{x}_1 = -0.38$ and -1.33 for constant margins. Without ϵ , the state reaches $\bar{x}_1 = 0.59$ but with 12.2% violation.

Table S14: Sparse GP Ablation: Position Constraint with Partial GP Coverage ($m = 3$, nonlinear perturbation, 5 seeds $\times$ 10 episodes $\times$ 200 steps). GP trained on $n=50$ samples in $x_1 \in [-0.3, 0.5]$; constraint at $x_{\text{lim}} = 1.2$. Results: mean $\pm$ std over 5 seeds.

ϵ Config.	Viol. (%)	CBF Viol. (%)	QP Infeas. (%)	$\bar{x}_1$	x_1^{max}	Mean ϵ
Compositional $\epsilon(x)$	0.00	0.00	0.0	0.37	0.74	0.15
ϵ_0^{mean}	0.00	0.00	0.0	-0.38	0.00	1.20
ϵ_0^{max}	0.00	0.00	0.2	-1.33	0.00	2.72
No ϵ ($\epsilon = 0$)	12.2 ± 0.0	56.8 ± 0.7	0.0	0.59	1.35	0

S13 Proofs of Theoretical Results

S13.1 Residual Gradient Bound

Lemma 1 (Residual gradient bound). *Let $\delta_i = \psi_i - \hat{\psi}_i^0$ with $\delta_0 \equiv 0$. Under the smoothness and gradient boundedness conditions stated in Theorem 1 of the main manuscript and smoothness of $\hat{f}$ on the compact operating region $\mathcal{X}_{\text{op}}$, $\|\nabla_x \delta_i(x)\| \leq G_\delta^{(i)}$ for all $x \in \mathcal{X}_{\text{op}}$, with*

$$\begin{aligned} G_\delta^{(0)} &= 0, \\ G_\delta^{(i)} &= H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f + (L_{\hat{f},i} + k_i) G_\delta^{(i-1)} + H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f \end{aligned} \quad (\text{S1})$$

where $H_\psi^{(i-1)} = \sup_x \|\nabla_x^2 \hat{\psi}_{i-1}^0\|_{\text{op}}$, $G_\psi^{(i-1)} = \sup_x \|\nabla_x \hat{\psi}_{i-1}^0\|$, and G_f is the gradient bound from the gradient boundedness condition of Theorem 1 of the main manuscript.

Proof. Differentiate δ_i term by term: $\delta_i = L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i) \delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1}$.

(i) $\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0) = \nabla_x^2 \hat{\psi}_{i-1}^0 \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \hat{\psi}_{i-1}^0$. By Cauchy-Schwarz and the gradient boundedness condition of Theorem 1 of the main manuscript: $\|\nabla_x (L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0)\| \leq H_\psi^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\psi^{(i-1)} G_f$.

(ii) $\nabla_x ((L_{\hat{f}} + k_i) \delta_{i-1})$ contributes $(L_{\hat{f},i} + k_i) G_\delta^{(i-1)}$.

(iii) $\nabla_x (L_{\Delta \hat{f}} \delta_{i-1}) = \nabla_x^2 \delta_{i-1} \Delta \hat{f} + (\nabla_x \Delta \hat{f})^\top \nabla_x \delta_{i-1}$, bounded by $H_\delta^{(i-1)} \|\Delta \hat{f}\|_\infty + G_\delta^{(i-1)} G_f$.

Summing yields Eq. (S1). Under the gradient boundedness condition of Theorem 1 of the main manuscript, $G_f = L_{\hat{f}} \|\Delta \hat{f}\|_\infty = \mathcal{O}(\|\Delta \hat{f}\|)$. By induction, $G_\delta^{(i)} = \mathcal{O}(\|\Delta \hat{f}\|)$ at every level, so $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \|\Delta \hat{f}\| = \mathcal{O}(\|\Delta \hat{f}\|^2) = \mathcal{O}(\rho^2)$. $\square$

S13.2 Full Proof of Theorem 1

Proof of Theorem 1. Step 1 (GP calibration). By the RKHS boundedness and GP calibration conditions stated in Theorem 1 of the main manuscript and the GP-UCB concentration inequality of Srinivas et al. [4], $|\Delta f_j(x) - \mu_{\text{GP},j}(x)| \leq \beta \cdot \sigma_{\text{GP},j}(x)$ for all $x \in \mathcal{X}$ holds with probability $\geq 1 - \delta/n$ per dimension. By union bound over $j = 1, \dots, n$, this holds simultaneously with probability $\geq 1 - \delta$. Condition on this event.

Step 2 (Compositional bound). We show $|\delta_i(x)| \leq \sigma_i(x)$ by induction on i .

- *Base case ($i = 1$):* $\delta_1 = \sum_j (\partial h / \partial x_j) \Delta \hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \leq \beta \sum_j |\partial h / \partial x_j| \sigma_{\text{GP},j} = \sigma_1$.
- *Inductive step ($i \geq 2$):* Assume $|\delta_{i-1}| \leq \sigma_{i-1}$. From Eq. (5) of the main text: $|\delta_i| \leq |L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0| + (|L_{\hat{f}}|_{\text{op}} + k_i) |\delta_{i-1}| + |L_{\Delta \hat{f}} \delta_{i-1}|$. Term 1 bounded identically to base case; Term 2 bounded by induction hypothesis; Term 3 bounded using Lemma S1: $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \beta \|\sigma_{\text{GP}}\|_2 = \sigma_{\text{cross}}^{(i)}$. Summing: $|\delta_i| \leq \sigma_i$.

The cross term is retained at full weight. Under the gradient boundedness condition of Theorem 1 of the main manuscript, it is formally $\mathcal{O}(\rho^2)$.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound Eq. (S2) yields $|\Delta(x, u)| \leq \sigma_m(x) + \sum_{j=1}^{m-1} c_j \sigma_j(x) + \sigma_{\text{ctrl}}(x) = \sigma_{\text{total}}(x)$. For $\kappa_\epsilon = 1$, $\epsilon(x) = \sigma_{\text{total}}(x)$, hence $|\Delta(x, u)| \leq \epsilon(x)$.

Step 4 (Forward invariance). For u^* satisfying $A_0 u^* \leq b_0 - \epsilon$: $\psi_m^{\text{true}}(x, u^*) = \hat{\psi}_m^0(x, u^*) + \Delta(x, u^*) \geq \epsilon(x) - |\Delta(x, u^*)| \geq 0$. By Theorem 2 of Xiao et al. [2], the safe set $\mathcal{C}$ is forward invariant. $\square$

S13.3 Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\text{ctrl}}(x) = \beta \sum_j \left| \frac{\partial(L_g L_{\hat{f}}^{m-1} h)}{\partial x_j} \right| \sigma_{\text{GP},j}(x) \cdot u_{\max} \quad (\text{S2})$$

where $u_{\max} = \sup_{u \in \mathcal{U}} \|u\|$. On the CCS, $\sigma_{\text{ctrl}}/\sigma_{\text{total}} < 0.15$ for both constraints.

References